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i **ASSIST does not take the place of a counselor on your campus.** It is intended to help students and counselors work together to establish an appropriate path toward transferring from a public California community college to a California university.

Major Articulation Agreement

Computer Science and Engineering/B.S.

Effective during the **2025-2026** academic year

	To: University of California, Los Angeles 2025-2026 General Catalog, Quarter	←	From: De Anza College 2025-2026 General Catalog, Quarter
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IMPORTANT MAJOR DETAILS

The Computer Science and Engineering curriculum provides students with the education and training necessary to design, implement, test, and utilize the hardware and software of digital computers and digital systems. The curriculum has components spanning from both the Computer Science and Electrical and Computer Engineering departments.

Admission to the Henry Samueli School of Engineering and Applied Science (HSSEAS) at UCLA is highly competitive. The most important selection criteria are completion of the required preparatory courses and academic performance. Listed below are the lower division preparation courses for the major. All applicants must have a minimum transferable cumulative GPA of 3.4. Preparatory courses should be completed by the end of the spring prior to Fall enrollment. HSSEAS admits students by major and does NOT consider applicants for alternate majors.

To be considered for this major, you must complete the following courses by the end of spring before transfer:

- One and a half years of Calculus through Multivariable,
- One course in Linear Algebra,
- One and a half years of Calculus Based Physics series [for Engineering majors] with Lab, and
- One course in C++ (strongly preferred), C or JAVA.

All courses must be taken for a letter grade.

Applicants are not required to complete the HSSEAS General Education Requirements in order to be admitted, although it is beneficial for students to complete 1 course from each of the following areas: arts, humanities, social sciences, and life sciences. Applicants can fulfill the lower division General Education requirements by completion of the Intersegmental General Education Transfer Curriculum (IGETC). Partial IGETC is NOT accepted.

For more information regarding this major and UCLA's transfer selection process, visit <https://admission.ucla.edu>. If you still have specific questions, you may email the HSSEAS admissions office at ugradadm@seas.ucla.edu.

PLEASE NOTE: The community college courses listed on this major agreement have been approved as substitutes to satisfy the admission preparation requirements for this major, but they may not be exact equivalents of the UCLA courses listed. In addition, upper division requirements for the major may be satisfied by lower division community college course(s) listed below, however, credit will be determined by the department after transfer.

PROGRAMMING REQUIREMENT

C++ is the PREFERRED LANGUAGE for this major, however (Java, & C) are also acceptable programming courses.

NOTE: A course equivalent to UCLA's CS 31 is acceptable to meet the programming requirement for this major.

AP EXAM CREDIT

Credit awarded by UCLA as a result of AP exams for students in the School of Engineering may be found here: <https://admission.ucla.edu/admitted-students/ap-credit-engineering>

LOWER DIVISION MAJOR REQUIREMENTS

1

Complete the following

A

COM SCI 31 Introduction to Computer Science I 4.00	Preferred courses CIS 22B Intermediate Programming Methodologies in C++ 4.50 OR CIS 22BH Intermediate Programming Methodologies in C++ - HONORS 4.50 ← OR CIS 29 Advanced C++ Programming 4.50 OR CIS 27 Programming in C++ for C/Java Programmers 4.50
COM SCI 32 Introduction to Computer Science II 4.00	CIS 22C Data Abstraction and Structures 4.50 OR CIS 22CH Data Abstraction and Structures - HONORS 4.50 AND Select 1 course from the following CIS 22A Beginning Programming Methodologies in C++ 4.50 ← OR CIS 22B Intermediate Programming Methodologies in C++ 4.50 OR CIS 22BH Intermediate Programming Methodologies in C++ - HONORS 4.50 OR CIS 27 Programming in C++ for C/Java Programmers 4.50 OR CIS 29 Advanced C++ Programming 4.50
COM SCI 33 Introduction to Computer Organization 5.00	← CIS 21JA Introduction to x86 Processor Assembly Language and Computer Architecture 4.50
COM SCI 35L Software Construction Laboratory 4.00	← No Course Articulated
EC ENGR 3 Introduction to Electrical Engineering 4.00	← No Course Articulated

2 Complete 1 course from A

A

COM SCI M51A Logic Design of Digital Systems	4.00	← No Course Articulated
EC ENGR M16 Logic Design of Digital Systems	4.00	← No Course Articulated
Same as COM SCI M51A		

3 Complete the following

A

MATH 31A Differential and Integral Calculus	4.00	← OR MATH 1ACalculus I5.00 MATH 1AHCalculus I - HONORS5.00
MATH 31B Integration and Infinite Series	4.00	MATH 1BCalculus II5.00 AND MATH 1CCalculus III5.00 OR MATH 1BCalculus II5.00 AND MATH 1CHCalculus III - HONORS5.00 ← OR MATH 1BHCalculus II - HONORS5.00 AND MATH 1CCalculus III5.00 OR MATH 1BHCalculus II - HONORS5.00 AND MATH 1CHCalculus III - HONORS5.00
MATH 32A Calculus of Several Variables	4.00	← OR MATH 1CCalculus III5.00 MATH 1CHCalculus III - HONORS5.00
MATH 32B Calculus of Several Variables	4.00	← OR MATH 1DCalculus IV5.00

			MATH 1DH Calculus IV - HONORS	5.00
MATH 33A Linear Algebra and Applications	4.00	←OR	MATH 2B Linear Algebra	5.00
			MATH 2BH Linear Algebra - HONORS	5.00
MATH 33B Differential Equations	4.00	←OR	MATH 2A Differential Equations	5.00
			MATH 2AH Differential Equations - HONORS	5.00
MATH 61 Introduction to Discrete Structures	4.00	←OR	MATH 22 Discrete Mathematics	5.00
			MATH 22H Discrete Mathematics - HONORS	5.00

4 Complete the following

A

PHYSICS 1A Physics for Scientists and Engineers: Mechanics	5.00			
AND				
PHYSICS 1B Physics for Scientists and Engineers: Oscillations, Waves, Electric and Magnetic Fields	5.00		PHYS 4A Physics for Scientists and Engineers: Mechanics	6.00
AND			AND	
PHYSICS 1C Physics for Scientists and Engineers: Electrodynamics, Optics, and Special Relativity	5.00	←	PHYS 4B Physics for Scientists and Engineers: Electricity and Magnetism	6.00
AND			AND	
PHYSICS 4AL Physics Laboratory for Scientists and Engineers: Mechanics	2.00		PHYS 4C Physics for Scientists and Engineers: Fluids, Waves, Optics and Thermodynamics	6.00
AND				
PHYSICS 4BL Physics Laboratory for Scientists and Engineers: Electricity and Magnetism	2.00			

5 Complete 1 course from A and B

A

ENGCOMP 3 English Composition, Rhetoric, and Language	5.00	←OR	ENGL C1000 Academic Reading and Writing	5.00
			ENGL C1000H Academic Reading and Writing - HONORS	5.00

AND

B

English composition courses

- COMM 9

Argumentation: Analysis of Oral and Written Communication

5.00
- OR
- EWRT 1B

Reading, Writing and Research

5.00
- OR
- EWRT 1BH

Reading, Writing and Research - HONORS

5.00
- ← OR
- EWRT 1C

Literature and Composition

5.00
- OR
- ENGL C1001

Critical Thinking and Writing

5.00
- OR
- PHIL 3

Critical Thinking and Writing

5.00

6 Complete 1 course from A

A

Computer programming courses: C++ preferred

- CIS 27

Programming in C++ for C/Java Programmers

4.50
- OR
- CIS 29

Advanced C++ Programming

4.50
- OR
- CIS 22A

Beginning Programming Methodologies in C++

4.50
- ← OR
- CIS 22B

Intermediate Programming Methodologies in C++

4.50
- OR
- CIS 22BH

Intermediate Programming Methodologies in C++ - HONORS

4.50

END OF AGREEMENT